ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Naval Warfare Officer (NWO) (00207)

Basic Arctic Navigation (BAN) (ALIB)

MITE Course IDs: 122173

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Training Development Centre (NTDC). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

Revision	Approval Date	Authority
1.0 (QSPWB Apr 2018)		NTDC(P) CO
		CNPTG

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FOREWORD

1. This training document is developed under the authority of the Commander Naval Personnel and Training Group (CNPTG) and in accordance with the following references:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-003/PQ-001 Basic Arctic and Offshore Patrol Vessel Navigation Specialty Specification dated 21 Feb 18;

   c. NAVORD 4500-0 “Structured Delegation”; and

   d. NPTORD 4500-22 “Qualification Standard and Plan Production and Management”.

2. This QSP consists of specific Performance Objectives (POs), Enabling Objectives (EOs), and Lesson Specifications in Chapter 3. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA (NTDC(P) Command and Leadership div), who will forward to NPTG for approval, as appropriate.

5. The designated TEs for this training are indicated below:

TE	Phase / Module	MITE Course ID	NQual
NFS(P)	All	122173	ALIB
NFS(A)	All	122173	ALIB

CHAPTER 1 TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for BWKs identified to serve aboard AOPVs and other platforms operating in the arctic. This QSP is designed to assist training establishments in developing, conducting, evaluating and validating the training.

2. The content within this QSP is based in the Face-to-Face (F2F) delivery of this course. The DLN content for this course will be the same as the F2F method, but will not require separate administration.

COURSE DURATION

3. Course duration is calculated in accordance with NPTORD 4500-22 Annex B. Detailed duration with resource allocation is found in Annex G of this QSP.

Allocation of ATT	**Minutes**
Sum of all EOs/POs	1400
Sum of all ECs	375
Sum of all PCs	510
Actual Training Time (ATT) Sub-total	**2285**

RESOURCES

4. The following table details key personnel required to conduct the training. Personnel are to be scheduled IAW the recommended course scheduling table in Annex G of this QSP.

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
Div Cdr	LCdr	NWO (MOSID 00207)	1	All
Instructor	Lt(N)	NWO (MOSID 00207) D-Level qualification (FNO preferred) DLI Qualification	1	All

5. The following table details the critical training resources to conduct the training. It should be read in conjunction with the recommended course scheduling table in Annex G of this QSP.

Critical Training Resources	PO/EO
DLN Account	All
Internet access	All

COURSE CAPACITY

6. Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (Instructor to Learner Ratios). Justification for course capacity decision can be found in Annex C.

   a. Maximum – 24; and

   b. Minimum – 1.

MITE

7. Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

8. Scheduling. Each session of this course is to be scheduled in MITE by the NFS conducting the training and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

9. Prerequisites. Priority for course loading is for personnel posted to an AOPV or a vessel bound for the arctic or operations in ice covered waters ($\geq$ 4/10th concentration). The loading authority must ensure that trainees have the NWO Phase IV before being selected for this course.

10. Qualification. Successful completion of this course constitutes the primary means of attaining the Basic Arctic Navigation qualification (ALIB; 122173).

COURSE CERTIFICATE

11. Upon passing this course, trainees shall receive a Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 “Course Reporting and Certificates” and NPTORD 4500-51 “Course Reports and Certification”.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

12. PLARs shall be conducted in accordance with NAVORD 4500-2 “RCN Prior Learning Assessment and Recognition”, DAOD 5031-1 “Canadian Forces Military Equivalencies Program”, and CFITES Vol 12 “Canadian Forces Military Equivalencies Programme: Prior Learning Assessment”.

CHAPTER 2 ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE NAVAL TRAINING DEVELOPMENT CENTRE (PACIFIC)

GENERAL

1. This is a pass/fail course. Trainees must successfully complete all POs to pass this course. Performance Checks (PCs) and Enabling Checks (ECs) are based on the Performance Objectives (POs) and Enabling Objectives (EOs) in Chapter 3 of this QSP. All PCs must be successfully completed to achieve the qualification. This QSP is based on the F2F delivery method. A trainee will be required to meet the same standard if they complete the DLN version of this course.

2. Trainees will be developed and assessed using a combination of formative and summative assessment events. At a minimum, these will involve PCs, ECs, end of lesson confirmation, homework, assignments, or quizzes. See Annex K for details.

3. Dress and deportment, ethics, conduct and behaviour demonstrated by the trainee shall be evaluated and assessed based on the standards laid out under Operation Honour and the LRH program. A trainee who displays inappropriate dress and deportment, ethics, and/or behaviour is considered to have demonstrated unsatisfactory conduct which may result in removal from training.

ENABLING AND PERFORMANCE CHECKS

4. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning to improve retention of course content. Formative assessments are designed to provide feedback to both instructional staff and trainees. Each EO must be formatively assessed prior to summative assessments.

5. In the event of an EC failure, the trainee will be counselled and given remedial training or additional homework, as appropriate.

6. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job.

7. PCs will be conducted after formative assessment has occurred and the trainee has had the opportunity to develop skills and apply knowledge related to the PO.

8. IAW NPTORD 4500-54 “Training Review Board Process” para 4.2, TRBs shall not be conducted for specialty training therefore no TRBs will be conducted for the BAN specialty qualification.

RESPONSIBILITY FOR TESTING

9. Assessment of trainees is to be done IAW CFITES Vol. 7 “Evaluation of Leaners”.

10. NFS staff provides quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NTDC(P).

PROGRESS MONITORING

11. Trainee progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning, so that content is retained.

12. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 “Conduct of Instruction Programmes”.

13. Continuous monitoring of candidate progress is required to provide the following:

    a. Early warning of trainee difficulties; and

    b. Feedback on the effectiveness of training.

14. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.

CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE NAVAL TRAINING DEVELOPMENT CENTRE (PACIFIC)

PO 001 Assess Impact of Environmental Conditions in Ice and at Northern Latitudes

1. Performance. Assess Impact of Environmental Conditions in Ice and at Northern Latitudes (T005).

2. Conditions.

   a. Given:

      (1) Operational HMC Ship's bridge;

      (2) Ice products;

      (3) Meteorological products;

      (4) Regulatory documents;

      (5) Internet access;

      (6) Support from CAF elements;

      (7) Assistance from OGDs; and

      (8) References.

   b. Denied: Nil.

   c. Environmental:

      (1) Onboard HMC ship; and

      (2) Any environmental conditions.

3. Standard. The BAN graduate shall assess the impact of environmental conditions in ice and at northern latitudes IAW refs by:

   a. (High) Assessing current ice conditions (T) by:

      (1) Analyzing signs of ice in the vicinity of the ship (T016) to include movement of the sea measurements based on observations (S);

      (2) Identifying ice characteristics (T001);

      (3) Interpreting information from CIS sources (T) by:

         (a) Obtaining current ice information (T);

         (b) Interpreting ice charts (T002); and

         (c) Measuring movement of sea ice based on ice charts (T);

      (4) Characterizing ice regime for specific location and vessel (T003) to include ice numerals (S001) calculation by ice eggs decoding (S); and

      (5) Reporting ice observations (T);

   b. (Med) Interpreting meteorological data for the Canadian arctic (T004); and

   c. (Med) Applying acts and regulations that govern arctic operations (T).

4. References.

5. Task Numbers. T001; T002; T003; T004; T005; T016; T024.

6. Competencies. Spatial orientation; Decision making; Working memory; Task prioritization.

7. Training Limitations.

   a. Training will be conducted with no actual exposure to ice or northern latitude environmental conditions.

   b. Instructors may not have significant ice experience.

   c. Depending on time of year, it may not be possible to source current ice products from appropriate CIS database.

PO 001 Assess Impact of Environmental Conditions in Ice and at Northern Latitudes

EO 001.01 Apply Acts and Regulations that Govern Arctic Operations

1. Performance. Apply Acts and Regulations that Govern Arctic Operations (T).

2. Conditions.

   a. Given:

      (1) Computer with access to DWAN / Internet; and

      (2) References.

   b. Denied: Nil.

   c. Environmental: DLN LMS.

3. Standard. The trainee shall apply acts and regulations that govern arctic operations IAW refs by:

   a. (Med) Explaining international regulations that pertain to ships operating in polar waters to include:

      (1) IMO Polar Code;

      (2) SOLAS;

      (3) MARPOL;

      (4) SCTW;

      (5) Guidelines For Ships Operating in Polar Waters; and

      (6) IACS;

   b. (Med) Explaining Canadian regulations that pertain to ships operating in polar waters by:

      (1) Interpreting AWPPA;

      (2) Applying ASPPR (T024);

      (3) Identifying how AIRSS applies to vessels; and

      (4) Identifying areas of enhanced environmental sensitivity; and

   c. (Low) Explaining roles and responsibilities of OGDs in the arctic to include:

      (1) MCTS;

      (2) CCG; and

      (3) ECCC.

4. Time Required.

   a. Nil – Trainees will progress DLN training at their own pace.

5. Remarks.

a. Trainees should be given electronic access to all references wherever possible.

b. Instructors may add scenarios or first-hand accounts.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Explain international regulations that pertain to ships operating in polar waters			
1. Identify the main structure and content in the IMO Polar Code 2. Identify the additional SOLAS requirements based on the polar environment 3. Identify the additional MARPOL requirements based on the polar environment 4. Identify the additional SCTW requirements based on the polar environment 5. Identify the general content and structure in Guidelines For Ships Operating In Polar Waters 6. Identify IACS requirements concerning polar classifications of ships		**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be given pictures, diagrams, videos at PPT presentations to walk through the various international regulations, highlighting critical elements specific to arctic operations. **Trg Resources:** • IMO Polar Code (for each trainee + instructor) • Guideline for Ship's Operating in Polar Waters (for each trainee + instructor) • MARPOL • SOLAS • SCTW excerpt for Ice Navigators (for each trainee + instructor)	
LESSON OBJECTIVE: Explain Canadian regulations that pertain to ships operating in polar waters			
1. Interpret the Arctic Water Pollution Prevention Act (AWPPA) 2. Apply ASPPR a. Identify how ASPPR applies to vessels based on classification (Canadian arctic class / type) b. Apply zone-date system i. Identify shipping safety control zones ii. Identify opening and closing dates for ships based on classification iii. Identify areas of no-entry c. Identify sewage, oily waste, and refuse deposit restrictions 3. Identify how AIRSS applies to vessels	AWPPA ASPRR AIRSS Ice Nav Manual	**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will walk through the various Canadian regulations, highlighting the sub-TPts. Trainees will demonstrate how to apply the zone-date system through scenarios which must be completed through DLN. **Trg Resources:** • References • Handouts of pertinent extracts for each trainee (if no electronic refs)	

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
4. Identify areas of enhanced environmental sensitivity			
LESSON OBJECTIVE: Explain roles and responsibilities of OGDs in the arctic			
1. Explain the MCTS System in the arctic to include: a. ECAREG Canada and reporting procedures b. MCTS zones c. NORDREG Canada and reporting procedures 2. Explain the roles and responsibilities of the CCG in the arctic to include: a. History of the organization in the arctic b. Coast Guard ice offices c. Icebreaker fleet i. icebreaker service fee ii. icebreaker service requests iii. escorting operations iv. icebreaker communication procedures d. CCG service and support in the arctic 3. Explain the roles and responsibilities of ECCC in the arctic to include: a. Relationship between METOC and ECCC b. Weather forecasts c. Weather charts d. Marine observations from vessels 4. Explain the roles and responsibilities of DND in the arctic to include: a. Strong, Secure, Engaged Defence Policy b. Canada's Northern Strategy c. Canadian Sovereignty	Ice Nav Manual Radio aids to marine nav	**Method / Media:** SME lecturers **Trg Environment:** Classroom / LMS **Activity Details:** Each OGD will have a SME lecturer by pre-recorded video who will run through the teaching points for their organization. Trainees will submit any questions to the instructor. The instructor will then be required to submit the question to the SME and provide the answer to the trainee(s). **Trg Resources:** • References	20 MCTS 40 CCG 20 ECCC 15 DND 30 Questi-ons

7. Test Details. EO 001.01 will be tested in EC 001.01 and PC 001 IAW table below and Annex K.

EC 001.01: Apply Acts and Regulations that Govern Arctic Operations

ASSESSMENT ACTIVITY: Trainees will have to complete a matching scenario by DLN LMS on their own time. ASPPR; AIRSS; Polar Code; AWPPA; Canadian Sovereignty issues; Canada's Northern

Strategy; Strong, Secure, Engaged Defence Policy; IMO Guidelines for Ship's Operating in Polar Waters; International Convention for the Prevention of Pollution from Ships to be covered at a minimum. The exam will be automatically corrected by the DLN LMS.
Pass/fail criteria: 70%

ASSESSMENT RESOURCES:

- References
- Computer with access to DLN LMS

IMPACT OF FAILURE: Remedial Training / Re-test on their own time

PO 001 Assess Impact of Environmental Conditions in Ice and at Northern Latitudes

EO 001.02 Assess Ice Conditions

1. Performance. Assess Ice Conditions (T).
2. Conditions.
   a. Given:
      (1) CIS products; and
      (2) References.
   b. Denied: Nil.
   c. Environmental: LMS.
3. Standard. The trainee shall assess ice conditions IAW refs by:
   a. (High) Identifying ice characteristics (T001) to include:
      (1) General ice terms;
      (2) Development of sea ice;
      (3) Types of ice;
      (4) Arrangement of ice;
      (5) Pack ice;
      (6) Ice surface features;
      (7) Stages of decay; and
      (8) Ice navigation terms;
   b. (High) Analyzing signs of ice in the vicinity of the ship (T016) by:
      (1) Detecting sea ice visually using the ice blink method;
      (2) Detecting open water visually;
      (3) Detecting sea ice by radar;
      (4) Measuring the movement of ice by observation (S); and
      (5) Explaining ice recce techniques;
   c. (High) Interpreting information from CIS sources (T) by:
      (1) Describing the different CIS products;
      (2) Obtaining current ice information (T);
      (3) Interpreting ice charts (T002);
      (4) Measuring movement of sea ice based on ice charts (T);
      (5) Recognizing changes in ice and weather conditions based on ice imagery;

(6) Characterizing the ice regime for a particular vessel (T003); and

(7) Analyzing principle tracking of icebergs in the arctic ( ); and

d. (Low) Reporting ice observations (T) by:

(1) Preparing an ice reporting message; and

(2) Preparing iceberg messages.

4. Time Required. Students to progress training at their own pace on DLN.

5. Remarks.

a. Depending on the time of year, it may not be possible to source current ice products from appropriate CIS database. In this case, the instructor can source archived information to achieve realistic training objectives.

b. If no current CIS products are available, students will be able to pull information from a mirrored CIS website with appropriate links build into the interface.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Identify ice characteristics			
1. Explain general ice terms to include; a. Polynyas; b. Ice aging; i. Salt content ii. Density/hardness iii. Change in colour c. Ice physics; i. Friction ii. Pressure iii. Drift based on current, wind, and floe depth d. Ice Properties i. Hardness ii. Salt content iii. Snow cover iv. Thickness 2. Explain the development of sea ice to include: a. New ice b. Nilas c. Young ice d. First-year ice e. Second year ice f. Multiyear ice g. Old ice	MANICE ECCC Ice Glossary Understanding and identifying old ice in summer	**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be shown pictures and videos of ice, as well as diagrams of ice growth and decay. The trainees will complete a DLN LMS homework assignment on completion of the lesson. The homework will include NLT 10 ice pictures to be matched with the appropriate terms and 10 short-answer questions on ice topics. **Trg Resources:** • Visual aids	

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
3. Explain different types of ice to include: a. Frazil ice b. Grease ice c. Slush d. Shuga e. Pancake ice f. Brash ice g. Growlers h. Bergy bits i. Icebergs j. Icecake k. Floe l. Floeberg m. Ice breccia n. Batture floes o. Fast ice p. Ice foot q. Anchor ice r. Grounded ice 4. Explain the arrangement of ice to include: a. Drift ice b. Pack ice c. Ice cover d. Concentration e. Ice distribution f. Openings in the ice g. Ice edge (extent) h. Ice boundary i. Iceberg limits 5. Explain pack ice to include: a. Open water b. Very open pack c. Open pack d. Close pack e. Very close pack f. Consolidated pack g. Compact pack h. Agglomerated brash 6. Explain ice surface features to include: a. Level ice b. Deformed ice c. Ice deformation processes d. Ice motion processes e. Ice ridges f. Hummocks g. Bummocks 7. Explain the stages of decay to include: a. Puddling b. Thaw holes c. Dried ice d. Rotten ice e. Flooded ice 8. Ice Navigation terms a. Beset b. Ice bound c. Nip d. Ice under pressure e. Iceports			
LESSON OBJECTIVE: Analyze signs of ice in vicinity of the ship			
1. Detect sea ice visually using ice blink method	MANICE	**Method / Media:** DL **Trg Environment:** LMS	

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
2. Detect open water visually using a. Water sky b. Frost smoke 3. Detect sea ice by radar a. Limitations of radar on detecting ice b. Differentiating ice from, land, aids to navigation and ships. 4. Measure movement of ice by observation a. Measuring movement by use of ARPA/Rel Vel/Trails b. Monitoring ice pressure against ship's hull c. Movement of pack ice d. Movement of Icebergs 5. Explain ice recce techniques to include: a. Use of helicopter b. Use of UAS c. Reports from vessels in the vicinity	Ice Navigation Manual	**Activity Details:** Trainees will be shown methods of analyzing signs of ice IVO the ship using visual aids. The trainees will be given pictures of building ice formations around the ship. **Trg Resources:** • Pictures • Videos • ECPINS Screen Captures	
LESSON OBJECTIVE: Interpret information from CIS sources			
1. Describing the different CIS products a. Regions Covered b. Types of charts i. Concentration chart ii. Stage of development chart c. WMO colour code table d. Symbol definitions e. Satellite Imagery f. Ice Berg Charts 2. Obtain Current Ice information a. Sourcing ice charts on CIS website b. Sourcing ice charts from foreign websites c. Sourcing Satellite Imagery d. Sourcing Ice Berg Charts 3. Interpret Ice Charts a. Explain the symbols used in ice charts to include i. dynamic process ii. crack iii. bergy water iv. strips and patches v. ice free vi. ice bergs	MANICE CH1 NOSTP 101 Canadian Ice Services	**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be given pictures, diagrams and virtual demonstrations to cover the TPs on ice services products and shown how to calculate ice numerals. The trainees will have to calculate NLT 5 ice numerals based on different ice charts and ship classes. **Trg Resources:** • Ice charts • Ice multiplier table	

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
vii. estimated ice berg limit viii. landfast ice ix. open water x. ice drift xi. openings in the ice xii. topographical features xiii. ice thickness xiv. stage of melting xv. snow cover xvi. ice of land origin xvii. defining limits xviii. coding for radar observations xix. colour coding b. Decode Ice Eggs i. Ice concentration ii. Coding for sea ice Stages of development iii. Coding for Lake Ice Stages of Development iv. Coding for form of Ice v. Coding and symbology for strips and patches vi. Coding for Brash 4. Measure movement of sea ice based on ice charts 5. Recognize changes in ice and weather conditions based on ice imagery 6. Characterize ice regime for particular vessel a. Identify purpose of AIRSS and ice regime b. Identify ship classification i. CAC/type vs Polar Class ii. AOPV ice classification c. Determine ice multipliers based on ship classification d. Calculate ice numerals e. Determine impact of positive/negative regime 7. Analyze principle tracking of ice bergs in the arctic.			
LESSON OBJECTIVE: Report ice observations			

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
1. Prepare ice reporting message to include but not limited to; a. Ice concentration b. Thickness c. Topography d. Ridges per linear mile e. Depth and surface coverage of snow f. Melt state g. Behaviour of ice i. Movement ii. Developing/releasing pressure h. Ridge heights i. Iceberg count j. Water temperature k. Remarks 2. Identify message addressees a. Action b. Info 3. Describing roles and responsibilities in reporting ice a. Ice observers b. Ice specialists 4. Preparing iceberg messages IAW MANICE CH4	MANICE CH2 MANICE CH4	**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be shown how to prepare an ice message using examples from old messages. Trainees will not be required to prepare a message, but will need to identify the different parts of the message. **Trg Resources:** • MANICE • Access to DLN LMS	15

7. Test Details. EO 001.02 will be tested by homework assignment and confirmation questions and assessed electronically during PC 001 IAW Annex K.

PO 001 ***Assess Impact of Environmental Conditions in Ice and at Northern Latitudes***

EO 001.03 Interpret Meteorological Data for the Canadian Arctic

1. Performance. Interpret Meteorological Data for the Canadian Arctic (T004).
2. Conditions.
   a. Given: References.
   b. Denied: Nil.
   c. Environmental: LMS.
3. Standard. The trainee shall interpret meteorological data for the Canadian arctic IAW refs to include:
   a. Freezing spray and super structure icing conditions;
   b. Precipitation and fog;
   c. Storm tracking in polar regions;
   d. Polar low pressure systems;
   e. Seasonal variability of sunlight;
   f. Arctic tides and currents; and
   g. Effects of polar latitudes on communications systems.
4. Time Required.
   a. EO – Trainees will progress DLN Training at their own pace.
   b. PC – There will be a time limit of 60 minutes for this PC.
5. Remarks: The SME lecturer is to be a forecaster qualified MET TECH (arctic/ice experience preferred).
6. Teaching Points.

Teaching Points	Ref	Training Activity Details	Time (mins)
LESSON OBJECTIVE: Explain arctic climatology and weather phenomena			
1. Describe freezing spray and super structure icing conditions to include: a. Ice accretion i. Ocean spray ii. Fog iii. Precipitation iv. Combination b. Freezing spray i. Contributing factors ii. Air temperature iii. SS temperature iv. Sea state	CF Weather office NOSTP 101 MANICE	**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be given pictures, diagrams and virtual demonstrations to cover the TPs. **Trg Resources:** PPT	80

Teaching Points	Ref	Training Activity Details	Time (mins)
v. Causes and effect 2. Describe precipitation / fog including water droplet surface tension 3. Explain storm tracking in polar regions to include: a. Summer months b. Winter months c. Air temperature patterns d. Precipitation patterns e. Atmospheric circulation 4. Describe effects of polar low pressure system to include: a. Polar lows b. Speed of formation and dissipation c. Wind speeds d. Basic requirement for development of formation 5. Explain seasonal variability of sunlight by latitude and month to include: a. Periods of extended daylight b. Periods of extended night 6. Describe arctic tides and currents to include: a. Transpolar drift b. Beaufort gyre c. Frobisher Bay tides 7. Describe effects of polar latitudes on communications systems to include: a. Katabatic outflow causing temperature inversions i. Ducting ii. Super-refraction b. Space weather and aurora			

7. Test Details. EO 001.03 will be tested in PC 001 IAW table below and Annex K.

PO 002 Perform Duties of OOW in Arctic / Ice covered Waters

1. Performance. Perform Duties of OOW in Arctic / Ice covered Waters (T012).

2. Conditions.

   a. Given:

      (1) Operational HMC Ship;

      (2) Bridge team;

      (3) Navigation aids;

      (4) Full chart outfit;

      (5) Ice products;

      (6) Meteorological products;

      (7) Regulatory documents;

      (8) Ancillary assets per mission requirements;

      (9) Support from CAF elements;

      (10) Assistance;

      (11) Supervision; and

      (12) References.

   b. Denied: Nil.

   c. Environmental:

      (1) Onboard HMC ship; and

      (2) Any environmental conditions.

3. Standard. The BAN graduate shall perform the duties of OOW in arctic / ice covered waters IAW refs by:

   a. (Low) Assessing the ship's position in ice covered waters and northern latitudes (T015) by:

      (1) Utilizing navigational charts for northern latitudes (T008);

      (2) Operating navigation systems in northern latitudes (T014); and

      (3) Interpreting navigation data in ice covered waters (T);

   b. (Med) Manoeuvring the ship in ice covered waters by:

      (1) Navigating the ship in the vicinity of ice;

      (2) Conducting ancillary asset operations in arctic/ice covered waters (T) to include:

         (a) Small boat operations;

         (b) Helicopter operations;

(c) MilCOTS operations;

(d) LOSV operations; and

(e) Mission payloads;

(3) Following ice leads (T018); and

(4) Executing sit and drift procedures (T); and

c. (Med) Managing engineering considerations for operations in the arctic (T010).

4. Task, Skill, Knowledge Numbers. T008; T010; T012; T014; T015; T018.

5. Training Limitations.

a. Instructors may not have significant ice experience;

b. Simulated Arctic environment may not be varied within the program

c. Simulated AOPV platform may not accurately reflect AOPV capabilities or interactions with ice;

d. Simulated ice characteristics may not be sophisticated enough to produce a realistic training environment; and

e. Inherent difficulties exist in simulating environmental conditions.

PO 002 Perform Duties of OOW in Arctic / Ice covered Waters

EO 002.01 Manage Engineering Considerations for Operations in the Arctic

1. Performance. Manage Engineering Considerations for Operations in the Arctic (T010).

2. Conditions.

   a. Given:

      (1) Assistance;

      (2) References

   b. Denied: Nil

   c. Environmental: LMS

3. Standard. The trainee shall manage engineering considerations for operations in the arctic IAW refs by describing AOPV ship design and construction to include:

   a. Hull design;

   b. Technical characteristics;

   c. Structural designs;

   d. Propulsion systems;

   e. Steering systems;

   f. Auxiliary systems;

   g. Torque requirements when operating in ice; and

   h. Stability considerations during icing conditions.

4. Time Required.

   a. Trainees will progress DLN at their own pace.

5. Remarks.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Describe AOPV ship design and construction			
1. Describe AOPV hull design a. Bow design b. Mid body shape c. Stern shape d. Ice belt 2. Describe AOPV technical characteristics a. Displacement and draft b. Length overall		**Method / Media:** DLN **Trg Environment:** LMS **Activity Details:** Trainees will be given pictures and diagrams of AOPV, or will utilize the virtual ship simulator if available. **Trg Resources:**	

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
c. Breadth d. Speed e. Range/Endurance f. Fin stabilizers g. Ice knives h. AOPV low temperature system performance 3. Describe AOPV structural designs a. Loading b. Structural arrangement c. Construction materials and behaviours at low temperatures 4. Describe AOPV propulsion systems a. Prime movers b. Electronic transmissions c. Mechanical transmissions d. Shafts e. Propellers f. Bow thruster g. Ice control modes 5. Monitor torque while operating in ice a. Entering/exiting ice b. Transiting in ice c. Engine thresholds 6. Describe AOPV steering systems a. Ice strengthened rudders b. Steering pumps c. Steering control modes and positions 7. Describe AOPV auxiliary systems a. Cooling systems b. Fuel systems c. Freezing of systems d. Waste disposal e. Heeling and Ballast tanks 8. Describe AOPV engineering requirements for operating in ice 9. Maintain AOPV ship's stability during icing conditions a. Preventative measures b. Corrective measures		• Canadian Virtual Naval Fleet • AOPS factsheet	

PO 002 Perform Duties of OOW in Arctic / Ice covered Waters

EO 002.02 Assess Ship's Position in Ice covered Waters and Northern Latitudes

1. Performance. Assess Ship's Position in Ice covered Waters and Northern Latitudes (T015).

2. Conditions.

   a. Given:

      (1) Screen Captures;

      (2) Canadian Arctic Charts;

      (3) Supervision;

      (4) Assistance; and

      (5) References.

   b. Denied: Nil

   c. Environmental: Classroom.

3. Standard. The trainee shall assess ship's position in ice covered waters and northern latitudes IAW refs by:

   a. (Med) Utilizing navigational charts for northern latitudes (T008) by:

      (1) Describing the shape of the Earth and approximations used in navigation;

      (2) Explaining chart projections;

      (3) Identifying chart datums; and

      (4) Explaining chart reliability;

   b. (Med) Operating navigation systems in northern latitudes (T014) by:

      (1) Utilizing aids to navigation in the arctic;

      (2) Explaining the effect of high latitude on gyros and compasses;

      (3) Explaining the effect of high latitude on electronic fixing aids; and

      (4) Explaining difficulties associated with fixing methods;

   c. (Med) Interpreting navigation data in ice covered waters (T) by:

      (1) Operating radar in the arctic; and

      (2) Identifying capabilities of unique AOPV navigation equipment.

4. Time Required.

   a. EO – Trainees will progress DLN at their own pace.

5. Remarks. Nil.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Utilize navigational charts for northern latitudes			
1. Describe the shape of the Earth and approximations used in navigation a. Shape of the Earth b. Use of datums c. Distances and their use d. Longitude/Latitude 2. Explain chart projections a. Requirements for navigational charts i. Correct distance scale ii. Bearing/courses iii. Rhumb line/straight lines iv. Accurate representation of land (equal area) v. Orthomorphism (correctness of shape) b. Preferred projections in the north i. Lambert conical projections ii. Polar stereographic iii. Mercator 3. Apply chart datums a. History of datums b. Geodetic system/ surveys c. GPS d. Ramifications of misuse e. Datum shifting 4. Explain chart reliability a. Date of survey/source data diagrams i. Current conditions may differ from charted features ii. Pingos b. Frequency and type of soundings i. Lead line ii. Single beam surveys iii. Multi beam full bottom coverage		**Method / Media:** DL **Trg Environment:** LMS **Activity Details: Trainees will be** given pictures and diagrams. Trainees will then be shown how to conduct a datum shift on ECPINS. **Trg Resources:** • PPT • Case Study • Canadian Arctic Charts (ENC and paper)	
LESSON OBJECTIVE: Operate navigation systems in northern latitudes			
5. Utilize aids to navigation in the arctic a. Limited availability b. RACON towers for PIs 6. Explain the effect of high latitude on gyros and compasses a. Magnetic		**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be given pictures and diagrams to present the teaching points. **Trg Resources:**	

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
b. Mechanical Gyro c. Ring Laser Gyro 7. Explain the effect of high latitude on electronic fixing aids a. ARPA information b. Echo Sounder c. GPS i. DGPS ii. Low angle iii. Effects of space weather d. Positional information on SINS 8. Explain difficulties associated with the following fixing methods: a. Dead Reckoning b. Astronomical Observations c. Visual Fixes		• PPT • Canadian Arctic Charts (ENC and paper)	
LESSON OBJECTIVE: Interpret navigation data in ice covered waters			
1. Operate radar in ice covered waters a. Identify primary method for fixing b. Discriminate between targets i. Ice ii. Vessels iii. Fixing aids c. Identify topography/fast ice i. Slow sloping shorelines ii. Lack of tidal data iii. Charting inaccuracies 2. Describe capabilities of navigation equipment unique to an AOPV a. Sigma 6 radar b. FLES		**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be given pictures and diagrams in addition the the PPT to explain the TPs. **Trg Resources:** • PPT • Canadian Arctic Charts (ENC and paper)	

7. Test Details. EO 002.02 will be tested in EC 002.02 and PC 002 IAW table below and Annex K.

EC 002.02: Assess Ship's Position in Ice covered Waters and Northern Latitudes

EC Time: 0 mins + 30 min review/debrief

ASSESSMENT ACTIVITY: The trainee will complete an assignment on LMS which will consist of constructing or interpreting an ice egg. This will be an open source assignment where students are expected to apply principles taught in this EO. Upon completion, the instructor will go over the standardized solutions while the trainees provide feedback where required.
Deficiencies will be addressed during instructor lead review. Remedial homework will be given for trainees who are having difficulty with the material.

ASSESSMENT RESOURCES:

- Chart Screenshots
- References

PO 002 Perform Duties of OOW in Arctic / Ice covered Waters

EO 002.03 Manoeuvre the Ship in Ice covered Waters

1. Performance. Manoeuvre the Ship in Ice covered Waters.
2. Conditions.
   a. Given:
      (1) Pictures;
      (2) Charts;
      (3) Scenarios;
      (4) Instructional videos;
      (5) Visual aids;
      (6) Supervision; and
      (7) References.
   b. Denied: Nil.
   c. Environmental: LMS
3. Standard. The trainee shall manoeuvre the ship in ice covered waters IAW refs by:
   a. (High) Navigating the ship in the vicinity of ice by:
      (1) Maintaining safe speed in presence of ice; and
      (2) Describing man-overboard considerations in ice.
   b. (High) Entering an ice floe under the CO's direction (T);
   c. (Low) Exiting an ice floe under the CO's direction (T);
   d. (High) Manoeuvring through an ice floe under the CO's direction (T) by:
      (1) Following ice leads (T018);
      (2) Protecting rudder and propellers;
      (3) Executing sit and drift procedures (T);
      (4) Employing techniques to avoid becoming beset (T020); and
      (5) Describing procedures to free own vessel from ice.
   e. (Low) Describing Seamanship evolution requirements in ice:
      (1) Describing fueling procedures in the Arctic;
      (2) Describing berthing procedures in ice covered waters;
      (3) Describing slipping procedures in ice covered waters; and
      (4) Describing anchoring in ice covered waters.
   f. (Med) Conducting ancillary asset operations in arctic/ice covered waters (T) to include:

(1) Small boat operations;

(2) Helicopter operations;

(3) UAS operations;

(4) MilCOT operations;

(5) LOSV operations; and

(6) Mission payloads.

4. Time Required.

   a. EO – Trainees to progress DLN at their own pace

5. Remarks. Nil.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Navigate ship in vicinity of ice			
1. Maintain safe speed in presence of ice a. Radar detection range i. Small contacts ii. Ice floes iii. Characteristics and efficiencies of radar b. State of visibility c. Power versus speed d. Safe speed in presence of ice bergs i. Ice berg limit ii. Area of uncertainty e. Appropriate distance from other vessels f. Soundings in ice i. FLES ii. Conventional echo sounder 2. Describe man-overboard considerations in ice	Ice Nav manual	**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be given visual aids to explain TPts . They will also be given a video with a voice over explaining what is going on. **Trg Resources:** • Visual aids • Videos	20 video
LESSON OBJECTIVE: Enter and exit ice floe under CO's direction			
1. Avoid entering ice whenever possible 2. Enter ice floe a. Hardness considerations b. Pressure considerations c. Thickness considerations d. Speed considerations i. initial impact ii. beaching iii. maintaining headway e. Best lead available	CCG Nav in ice covered waters Ch.4	**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be given visual aids to explain TPts . Trainees will also be given virtual demonstrations using voice overs. **Trg Resources:** • Visual aids • Video	15 video

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
f. Best angle to ice (90 degrees) g. Bow shear considerations h. Torque limits i. Entry point adjustments as required 3. Exit ice floe with considerations for power versus speed when entering open waters			
LESSON OBJECTIVE: Manoeuvre through ice floe under COs direction			
1. Follow ice leads a. Describe types of leads i. Line ii. Flawed iii. Open iv. Shore b. Choose path of least resistance c. Avoid glacial ice (bergs, bergy bits, growlers) d. Identify pressure ridges (hummocks, bummocks) i. How to avoid ii. Actions in case of building pressure (Stop until pressure event ended) 2. Protect rudder and propellers a. Centering rudder when backing away b. Milling c. Stalling d. Ice knife and Ice horn 3. Execute sit and drift procedures a. Monitor ice floe movement b. Monitor ship movement c. Watch for ice bergs d. Consider NFP requirements 4. Employ techniques to avoid becoming beset a. Star manoeuvre b. Backing and ramming 5. Describe procedures to free own vessel from ice a. Use of heeling tanks b. Use of fire hoses to reduce friction c. Use of propeller wash d. Request assistance		**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be given descriptions and virtual demonstrations to explain the TPs. Trainees will also be given virtual demonstrations using voice overs. **Trg Resources:** • Visual aids • Video	20 video

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Describe Seamanship evolution requirements in ice			
1. Describe fueling procedures in the Arctic a. NNF b. CCG c. Commercial barge d. Environmental considerations 2. Describe berthing procedures in ice covered waters a. Berth clear or foul i. Ice characteristics (brash vs fast) ii. Techniques to clear berth b. Jetty characteristics 3. Describe slipping procedures in ice covered waters a. Ice pressure b. Forward vs astern departure (based on ice conditions, current and moving ice) c. Flushing the stern of ice 4. Describe anchoring in ice covered waters a. Techniques: i. Conventional ii. Deadman's byte iii. In iceberg fields b. Considerations: i. Ice accretion on anchor cable, hawse pipe and ground tackle ii. Length of cable iii. Power to windlass iv. Density of ice (brash, rotten and scattered floes) v. Position of ice pack/floe vi. Engineering configuration (NFP) vii. Potential requirement to expedite weighing anchor		**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be given pictures, diagrams and virtual demonstrations to cover the TPs. **Trg Resources:** • Visual aids • Berthing and slipping animated videos (TEL)	
LESSON OBJECTIVE: Describe ancillary asset operations in arctic/ice covered waters			
1. Conduct small boat operations a. Launching/recovering b. OTH deployment i. Communications ii. Navigation		**Method / Media:** DL **Trg Environment:** LMS **Activity Details:** Trainees will be given pictures, diagrams and virtual demonstrations to cover the TPs.	

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
iii. Range / fuel c. Recce i. Ice leads ii. Observing tidal ranges d. Proceeding ashore i. Instructions to boat crew ii. Landing craft iii. Small boat e. Survival equipment f. Protective equipment (PPE, weapons, etc) 2. Conduct helicopter operations 3. Manage UAS operations 4. Manage MilCOT operations 5. Manage LOSV operations 6. Describe mission payloads		**Trg Resources:** • Visual aids • Animated videos (TEL)	

7. Test Details. EO 002.03 will be tested in EC 002.03 and PC 002 IAW table below and Annex K.

ANNEX A - *MAIN REFERENCES*

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

Ref Code	Publication
A1	RCN Arctic Operations Manual
A2	Command of an Arctic Offshore Patrol Vessel (AOPV)
A3	Harry DeWolf Class AOPS – CONUSE ver. 1.0
A4	CFCD 130 Canadian Navigation Manual ver.1.7
A5	MANICE Canadian Ice Services
A6	CJOC Plan for the North
A7	Arctic Waters Pollution Prevention Act
A8	CJOC Commander's Planning Guidance – Operation NANOOK 2018
A9	Operation NANOOK 2018 – Maritime Exercise Initial Planning Conference Administrative Instructions
A10	Canadian Forces Northern Employment and Support Plan, Nov. 2012
A11	Optimal RSOM-hub Locations for Northern Operations (Defence Research and Development Canada)
A12	MOU Between DND and DFO Concerning training of RCN personnel on Ships of the CCG operating North of 60 degrees
A13	MOU Between DFO/CHS) and DND regarding provision and distribution of hydrographic charts and publications, data, and mutual cooperation
A14	Ice Navigation in Canadian Waters (CCG)
A15	Canadian Ice Services (https://www.canada.ca/en/environment-climate-change/services/ice-forecasts-observations/latest-conditions.html)

Ref Code	Publication
A16	Northern Approaches – The Army Arctic Concept 2021
A17	Polar Continental Shelf Program – Arctic Operations Manual
A18	The Canadian Maritime Arctic and Northwest Passage
A19	Royal Canadian Navy Intelligence, Surveillance, Target Acquisition and Reconnaissance Unmanned Aircraft System (RCN ISTAR UAS) Project
A20	TP12259E Arctic Ice Regime Shipping System (AIRSS) Standards
A21	TP12819E User Assistance Package for the Implementation of Canada's Arctic Ice Regime Shipping System (AIRSS)
A22	TP14044E Arctic Ice Regime Shipping System Pictorial Guide
A23	TP4338E Command of an Ice Breaker Volume 3
A24	Tri Command Arctic Action Plan
A25	Arctic Operations and Northern Re-Supply
A26	Statement on Canada's Arctic Foreign Policy
A27	Nanisivik Naval Facility – Concept of Support to Operations
A28	HDW Class SEMS
A29	Ice Breaking Operations Directive 1: Provision of Ice Breaking Services (DFO/CCG)
A30	Ice Breaking Operations Directive 2: Commercial Harbour Breakouts (DFO/CCG)
A31	Ice Breaking Operations Directive 3: Fishing Harbour Breakouts(DFO/CCG)
A32	Ice Breaking Operations Directive 4: Support to Sealing Vessels (DFO/CCG)
A33	Hydrographic Services Handbook (HSO)
A34	Notices to Shipping – Central Canada and the Arctic
A35	Radio Aids to Marine Navigation – Atlantic, St. Lawrence, Great Lakes, Lake Winnipeg and Arctic
A36	Radio Aids to Marine Navigation – Pacific and Arctic

Ref Code	Publication
A37	Annual Edition of Notices to Mariners – Central and Arctic
A38	List of Lights, Buoys and Fog Signals (Vol. 1-4)
A39	The Canadian Aids to Navigation System
A40	CHS Canadian Tide and Current Tables (Vol. 1-4)
A41	CHS Nautical Charts and Publications (Catalogue 4 – Arctic)
A42	CHS Northern Canada Sailing Directions (ARC 400E – ARC 404E)
A43	CHS Atlantic Coast Sailing Directions (ATL 100E – ATL 121E)
A44	DAOD 8007-0 –Notification of DND and CF Activities Within the JTF(N) AOR
A45	MARLANT Standing Environmental Assessment for Arctic Operations
A46	Marine Communications and Traffic Services (MCTS) – Central & Arctic Region
A47	CIS Annual Arctic Ice Atlas
A48	MOU Between DND and DFO Concerning Fisheries Patrols and Surveillance
A49	MOU Between DND and Denmark Concerning Arctic Cooperation
A50	MOU Between DND and DFO Concerning HMC Ship Personnel as Fisheries Officer
A51	Arctic Voyage Planning Guide
A52	HMCS Harry DeWolf – Captain's Ship Book
A53	Mariner's Handbook
A54	Transport Canada Ships Safety Bulletin 01 2018: How to Meet STCW Requirements for Masters, Deck Officers and Other Crew Members of Certain Canadian Ships Operating in Polar Waters
B1	BR45(6) – Ice Navigation
B2	MODIS Worldview Imagery

Ref Code	Publication
B3	ATP 17 Chapter 10
C1	Ice Classed Ships: Main Engines (MAN)
C2	M303 – Arctic Offshore Patrol Ship (Odense Maritime Technology)
C3	WECDIS Operator's Manual
C4	FLES Operator's Manual
C5	Sigma 6 Operator's Manual
C6	Lloyd's Standard for Polar Class Ships
C7	American Shipbuilding Standards
D1	IMO Polar Code 01 January 2017
D2	IMO Guidelines for Ships Operating in Polar Waters
D3	Ice-Induced Loads on Ship Hulls (Norwegian University of Science and Technology)
D4	Arctic Marine Operations Report
D5	IACS Requirements Concerning Polar Class
D6	MARPOL – International Convention for the Prevention of Pollution from Ships Lloyd's Register Version 9.4
??	NOSTP 101

ANNEX B - *TASK LIST*

Task #	Task Statement	Train / No Train	N/T Reasoning	PO
Ice and northern latitudes				
T001	Identify ice characteristics	T		001
T002	Interpret ice charts	T		001
T003	*Characterize* ice regime for specific location and vessel	T		001
T004	Interpret meteorological data for Canadian Arctic	T		001
T005	Assess impact of environmental conditions *in ice and* at northern latitudes	T		001
T006	Calculate magnetic deviation and variation	NT	Previously trained NWO Ph trg	-
T007	Calculate tides and currents in ice	NT	Previously trained NWO Ph trg	-
T008	Utilize navigational charts for northern latitudes	T	Specific to chart projections/accuracy	002
	Obtain current ice information	T	From AAN but at lower level	001
	Measure movement of sea ice based on ice charts	T	From AAN but at lower level	001
	Report Ice Observations	T	From AAN but at lower level	001
	Interpret information from Canadian Ice Services sources	T	From AAN but at lower level	001
	Assess ice conditions	T		001
Vessel Performance				
T009	Take precautions for ship operations in ice and cold climate	NT	Experiential, KYSB, Previously trained	-
T010	Manage AOPV engineering considerations for operations in the Arctic	T		002
T011	Implement DC conditions for ice	NT	SSOs, experiential	-

Operating a ship in ice				
T012	Perform duties of OOW in *Arctic/ice covered waters*	T		002
T013	Apply shiphandling principles for ice covered waters	NT	Previously trained (no difference for ice)	-
T014	Operate navigation systems in polar latitudes	T		002
T015	Assess ship's position in ice covered waters and northern latitudes	T		002
T016	Analyze signs of ice in vicinity of the ship	T		001
T017	Manoeuvre an AOPV through open water leads	NT	Previously trained (no difference for ice)	-
T018	Follow ice leads	T		002
T019	Manage icing conditions on the ship	NT	Previously trained	-
T020	*Employ* techniques to avoid becoming beset in ice under CO's direction	T	Higher level task (AAN); Changed to reflect job requirements at BAN level	002
T022	Utilize communications equipment in polar latitudes	NT	Taught previously; different trade consideration	-
	Interpret navigation data in ice covered waters	T		002
	Manoeuvre ship in ice-covered waters	T		002
	Navigate ship in vicinity of ice	T		002
	Execute sit and drift procedures	T		002
	Enter an ice floe under CO's direction	T		002
	Exit an ice floe under CO's direction	T		002
	Conduct ancillary asset operations in Arctic/ice covered waters	T		002
Regulatory documents				
T024	Apply Arctic Shipping pollution prevention regulations	T		001

	Apply Acts and Regulations that Govern Arctic Operations	T		001
Environmental Factors and Regulations				
T026	Take precautions in arctic areas of enhanced environmental sensitivity	NT	Experiential; not a new task since actions stay the same (difference is knowledge piece)	-

ANNEX C - *QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS*

Dates: 23 Apr – 2 May 18

Location: NTDC(P), Work Point, CFB Esquimalt

Board Members:

Position	Rank and name	MOSID / Unit	Dates
Chair	*Cdr Corey Gleason*	*00207 / HMCS HARRY DE WOLF*	*Not present*
Chair	Lt(N) Mike Ronaldson	00207 / NTDC(P)	23 Apr – 2 May 18
SME	Lt(N) Landon Zeeman	00207 / NTDC(P)	24-26, 30 Apr, 1-2 May 18
SME	Lt(N) Tyler Stevens	00207 / NFR	23 Apr – 2 May 18
SME	WO Brian Fisher	00100 / Trinity METOC	23 Apr – 2 May 18
TDO	SLt Sarah Valentine	00211 / NTDC(P)	23 Apr – 2 May 18
Secretary	A/SLt Luke Brannigan	00207 / NTDC(P)	23 Apr – 2 May 18

Board Objective:

To review and update the Arctic Operations Course (119054) and ensure it is in line with the new NWO speciality specification. Any skills, knowledges or tasks identified on the Advanced Arctic Navigation board pertinent to basic arctic navigation were incorporated.

Discussions / Decisions:

**Decisions which are stricken through were changed at a later date*

Date	Decision/Event	Detail
23 April 2018	Welcome brief from the NTDC(P) DCO	
	Intro brief from SLt Valentine (TDO)	Layout of the board, timeline, CFITES overview
	Target audience	Discussion on whether the intent of the QSP is just for BWKs on AOPVs or every type of ship going to the arctic. Board decided the course was designed for any BWK identified to go into ice covered waters on any platform.
	Doc Review	• SPQR for NWO and Met Tech • AAN ROP as it relates to BAN • Existing AOC QSP PO Standard statement
		Job laid out as follows for NWO : • AOPV specific info only as required (eg. Engineering); generally platform non-specific (i.e. navigation) • ~~Where it fits into FNO course to be decided~~ (PLAR?) • To ensure the professional manoeuvring of Canadian Forces vessels, a requirement exists for a specialist officer to carry out the duties of BWK onboard an AOPV that is capable of conducting the

Date	Decision/Event	Detail
		full spectrum of military operations year round in medium first year ice, which may include old ice inclusions.
		Job laid out as follows for Met tech : • No ice forecasting • Forecasting weather/meteorology; Analyze and interpret arctic ice services products and weather data • Advising command on environmental conditions that affect ship's operations
	Train/no train task list	Board completed T/NT and compared to the current arctic operations course. NT decisions included at Annex B
	Grouping tasks	Board initially grouped tasks as follows: 1. Environmental conditions (ice and met) 2. Navigating 3. Canadian Arctic specific ~~4. Communicating with icebreakers/CCG~~
	Assessing weather and ice	When assessing weather and ice conditions, consider past, present, and future: Past, the data; present, what's happening now; and future, what's going to happen. Board decided this was a logical way of structuring the performance.
	~~Briefing command~~	~~Created task "Brief command on impact of environmental conditions at northern latitudes": It was decided that MET TECHs would brief environmental conditions and NWOs would brief the impact~~
	T012	Changed from "perform duties of an OOW in AOPV in the Arctic" to "perform duties of an OOW in Arctic/ice covered water". Changed because these duties may have to be done on other platforms, not just AOPV and would be affected not only in the Arctic but also in ice covered waters.
	T020	~~Changed to "Describe techniques to avoid becoming beset in ice".~~ Changed to "Employ techniques to avoid becoming beset in ice under the direction of a CO." Changed because an OOW will never be alone if the ship is in risk of becoming beset in ice, but they will be supporting those who have completed the AAN course (CO, XO, NavO) ~~so should understand what's going on~~.
	T021	Changed to "Describe procedures to free own vessel from ice" because it's at a lower level (same vocabulary) than the AAN course.
	T010	Removed AOPV from the task because it is engineering considerations in the Arctic, not just on AOPVs (platform non-specific).
24 April 2018	Completed task analysis	Scalars for 2 POs: PO 001 Assess the impact of environmental conditions in ice and northern latitudes PO 002 Perform the duties of OOW in arctic/ice covered waters
	PO writing brief	Given by TDO
	Job title: BAN Graduate	Board decided that "BAN Graduate" will be the title of someone who has completed the BAN course. Discussion: Some candidates may not be BWK qualified, therefore "Arctic BWK" won't work.
	T003	Changed verb from "calculate" to "characterize" IOT bring in line with the Ice Navigation in Canadian Waters reference.
	T025	"Apply IMO standards for operation in remote areas" has been downgraded to a skill instead of a task as it is something they will do continuously (not time-bound)
	T026	"Take precautions in arctic areas of enhanced environmental sensitivity" made "No Train" because they need to know about the precautions and the

Date	Decision/Event	Detail
		areas of enhanced environmental sensitivity but they are already familiar with the requirement to review references and sailing directions to determine environmental restrictions for a given area.
	Finished PO writing	
	SitRep	Board composed a SitRep to Cdr Gleason which included the following recommendations for his consideration:
		Met Tech. There are only four "train" tasks for Met Tech (highlighted in yellow above) and all of these are also "train" for NWO. Though we were considering having the course modularized for both Met Tech and NWO to take module 1 (equivalent to PO 001 Environmental conditions), this would involve Met Techs being retrained on several tasks and over-trained on others. The **recommendation** of the board is therefore to have the BAN course specific to NWOs and to create a self-study package encompassing the 4 trained tasks for Met Techs to cover off their SPQR.
		FNO. The board believes this course could better serve RCN needs as a standalone course. If this course is attached to FNO, some candidates may be over-trained or miss the "right time" piece for training (i.e. FNOs who are not going to the Arctic) and some who are going on deployment and require the training may not be able to get it because the course serials are filled with FNO trainees (limited instructor availability). The **recommendation** of the board is to have BAN as a fully standalone course, not associated with FNO, so instructor and candidate time is more effectively and efficiently managed (eg. Right training, right time, right place).
		Course prerequisites. The board believes this course is beneficial to all post-NWO phase IV who are proceeding to the arctic. The current course had a prerequisite of a bridge watchkeeping certificate. Board **recommendation** is to remove BWK certificate from prerequisites, recognizing there may be a need for loading prioritization as course capacity is dependent on how many NABS cubicles are fitted with ice modelling characteristics and equipment.
25 April 2018	Target Population	Completed the target population and TE analysis, giving the board a better understanding of the capabilities on each coast (eg. the lack of computer availability at NFS(A) classroom) and raised questions as to the course capacity. Board decided that there will be no more than 6 trainees per instructor.
	Instructional Analysis brief	Given by TDO to board.
	Instructional Analysis	Board added SPQR knowledge and skills to existing scalar, reconfiguring for instruction versus performance on the job when applicable. Board recognized that some teaching points will come directly from existing Arctic Operations Course (from FNO).
	- / T0012 Communicate with a CCG Ice Breaker	Decided by the board to be "no train" task as it was decided that NWOs and Met Techs only need to know about CCG capabilities (knowledge piece) but not communicate with them.
	Ice blink vs Water blink	Board discussed whether the correct terms were "ice blink" and "water sky" or "K011/K0010 water blink" and "K010/K0009 ice sky" (according to Shanghai university course). Internet search showed that "ice blink" and "water sky" were correct.
	Effects on crew	Board decided not to included effects on crew (i.e. snow blindness, hypothermia, etc.) because it is already taught on previous courses to look out for the crew and is collective training
	Writing Assessments brief	Given by TDO
	PC 001	Board decided that the performance associated with the PO would be giving a decision brief where all environmental factors were considered.

Date	Decision/Event	Detail
		Board decided that the process was more important than the product therefore trainees will not be assessed on their briefing skills or the COA they choose but will instead be assessed on the completeness of environmental factors considered and justifications given.
	TRB	IAW NPTORD 4500-54 "Training Review Board Process" para 4.2, TRBs shall not be conducted for specialty training therefore it is with board agreement that TRBs will not be held for BAN training.
26 April 2018	**Parking Lot review**	
	T022	Board decided that T022 is a N/T because general communication skills are taught on previous courses and the comms equipment affected by polar latitudes is a consideration of the Nav Comms and Ops more than NWOs.
	T023	Board decided to change task to a knowledge because NWOs are already familiar with reviewing radio aids to marine navigation to determine the MCTS requirements.
	T017	Board decided task was N/T because open water leads are the same as open water and NWOs are already trained on it
	Writing EOs Brief	Given by TDO
	EC 001.03	Board decided no EC for EO 001.03 (Meteorology) was required because the scope involves specific knowledge elements that will be assessed on the PC but are not different enough from previous meteorological exposure to deserve their own EC. The SME lecturer will be expected to ask confirmation questions to confirm trainee understanding.
27 April 2018	EC 001.02	Board decided there will be no EC for EO 001.02, instead they will be formally assessed throughout the EO and in PC 001
	Completed PO 001	Board completed PO 001
30 April 2018	K032	The board decided to "no train" emergency station procedures in ice because it is unclear what the procedures will be (they are not yet developed) or whether they will differ from existing procedures. This may be subject to change when more information on AOPV emergency procedures in ice become available.
	EO 002.01	TPts are specific to AOPV because all NWOs have the potential to serve aboard an AOPV and if NWOs on MCDVs take this course, it will save them from having to come back a second time if they are posted to an AOPV. The EC will include comparisons to other classes of ship so that they are informed of specifications outside AOPV as well.
01 May 2018	Course Resources (Staff)	When looking at course staffing in Chapter 1, board decided the limitation is 6 students per 1 instructor as that is the max per NABS cubicle
		For MET TECH SME lecturer, preferred WO but Sgt is acceptable
		Discussion about qualification level for an instructor. All agreed that the instructor would have to have BAN, however the board was unsure about FNO or AAN qualification. Because this is only basic navigation, an FNO instructor is likely not required. AAN (ALIF) would be the preferred instructor qualification however that will be difficult until the course is more established and limits the instructor to personnel who have been CO, XO, NavO on an AOPV. Arctic or ice covered waters experience would be preferred however that is difficult to monitor (no MITE/NQual associated with it). ~~It was decided by the board to make the instructor AAN qualified knowing that it will likely not be possible in the immediate future.~~
	Defining Ice terms used throughout the QSP	"Open water" = ice free
		"Vicinity of ice" = Ice within 6nm
		"Ice covered" = 1/10 or more; poses danger to the ship
		"Consolidated ice" = Defined as 10/10 concentration
	Chapter 1 and 2	Board completed Chapter 1 and 2. Timing table in Chapter 1 remains to be completed.

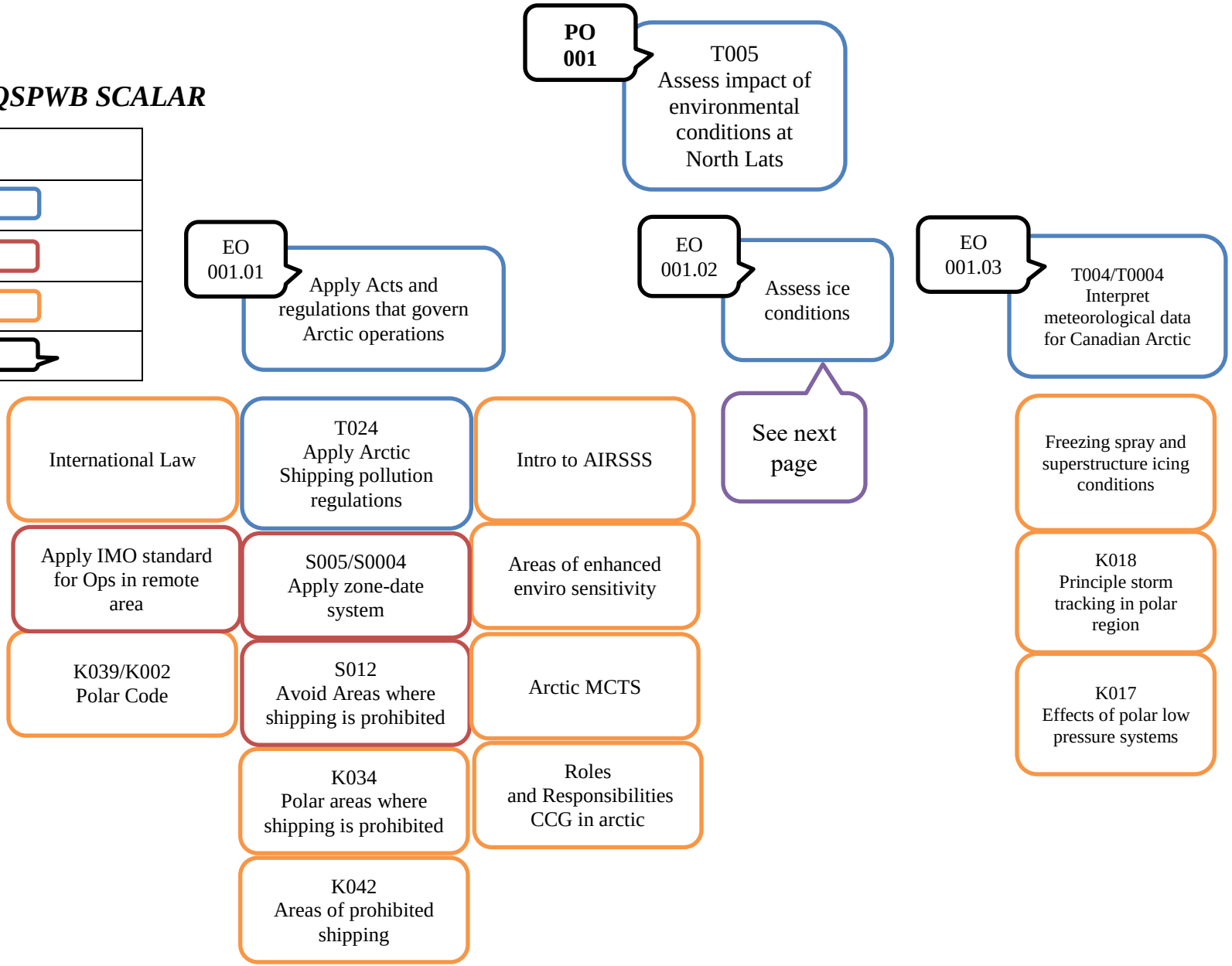
Date	Decision/Event	Detail
2 May 2018	**RoP Review**	
	Course instructor qualification	Discussion continued about qualifications for the instructor. It was discussed that AAN is too restrictive, BAN qualified would potentially be too junior, arctic experience would be preferred, FNO would be preferred but unnecessary. Board decided that the course instructor qualification should be BAN qualified with D-Level qualification (FNO preferred).
	FNO	Board decided BAN will be removed from FNO QSP and be its own standalone course because all FNOs won't require it. PO 007 of FNO will need to be removed from the current course (QSP amendment).
		For scheduling: On the west coast, it would be beneficial to have BAN scheduled immediately after the FNO course to facilitate FNO graduates who are posted to arctic-bound ships to take the course.
		For scheduling: Course should be scheduled in the Spring whenever possible, prior to the navigable season.
	Assistance vs support	In the context of Given Conditions in the POs, "Support" is mandated by an operational requirement (internal); "Assistance" is requested from outside agency.

Follow Up Action:

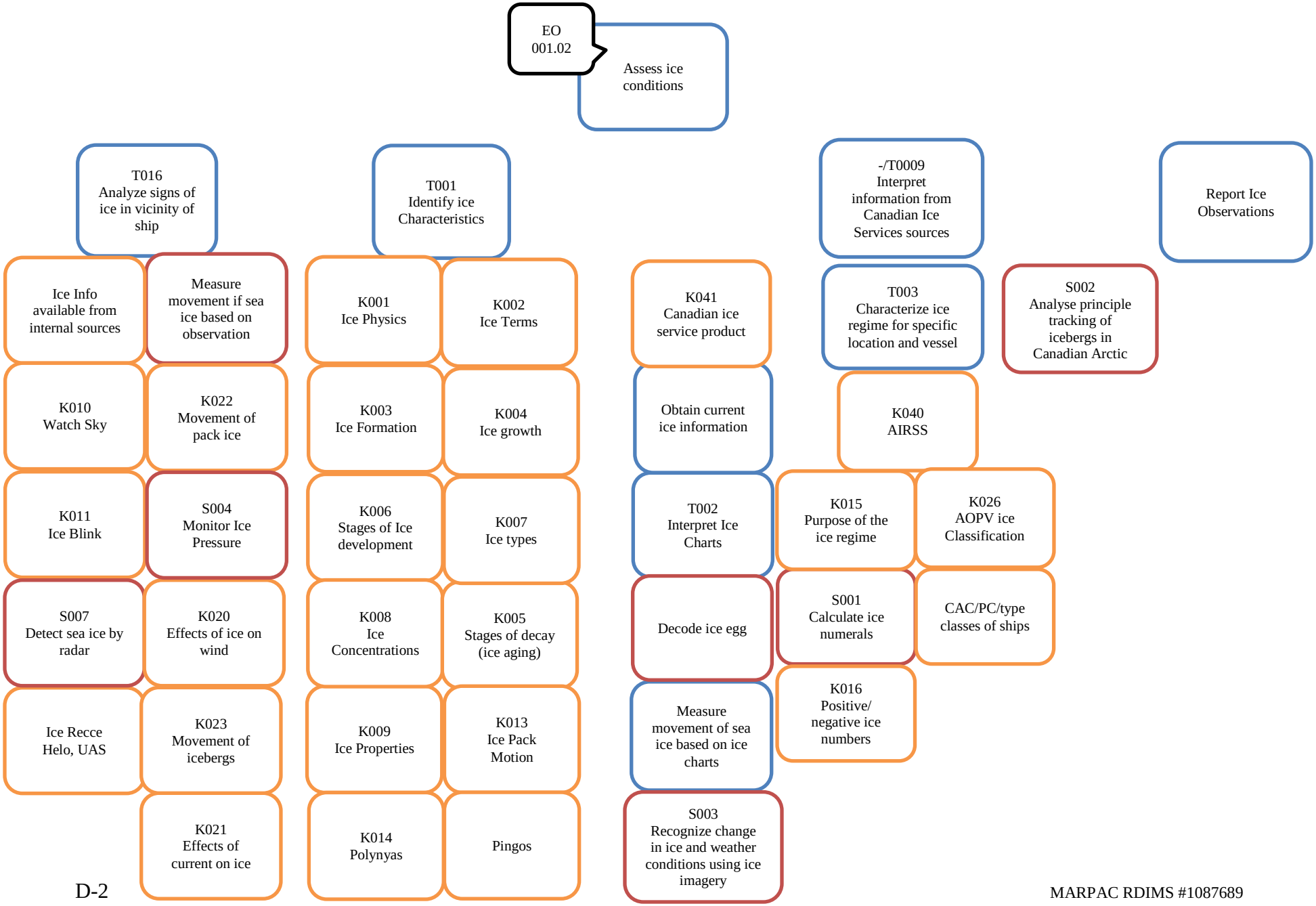
Item	Action	OPI	Expected Date of Completion
1	Board recommendations on Met Tech inclusion, FNO placement, and Course prerequisites endorsed by Cdr Gleason	Board Chair	11 May 18
2	QSP submitted for approval	NTDC(P) C&L Div Cdr	31 May 18
3	SPQR amendment submitted	NTDC(P) C&L Div Cdr	31 May 18
4	PO 007 (Arctic Operations Course MITE 119054) removed from FNO course by QSP amendment (*pending recommendation approval*)	NTDC(P) C&L Div Cdr	31 May 18
5	Learning Support Requests submitted for animated video creation	NTDC(P) C&L Div Cdr	29 Jun 18
6	Development of ECs and assignments to be converted to DLN LMS completed	NTDC(P) C&L Div Cdr	13 Jul 18
7	Learning Support Requests submitted for DLN LMS conversion material	NTDC(P) C&L Div Cdr	20 Jul 18
8	Course material fully developed	NTDC(P) C&L Div Cdr	31 Dec 18

ANNEX D - *QSPWB SCALAR*

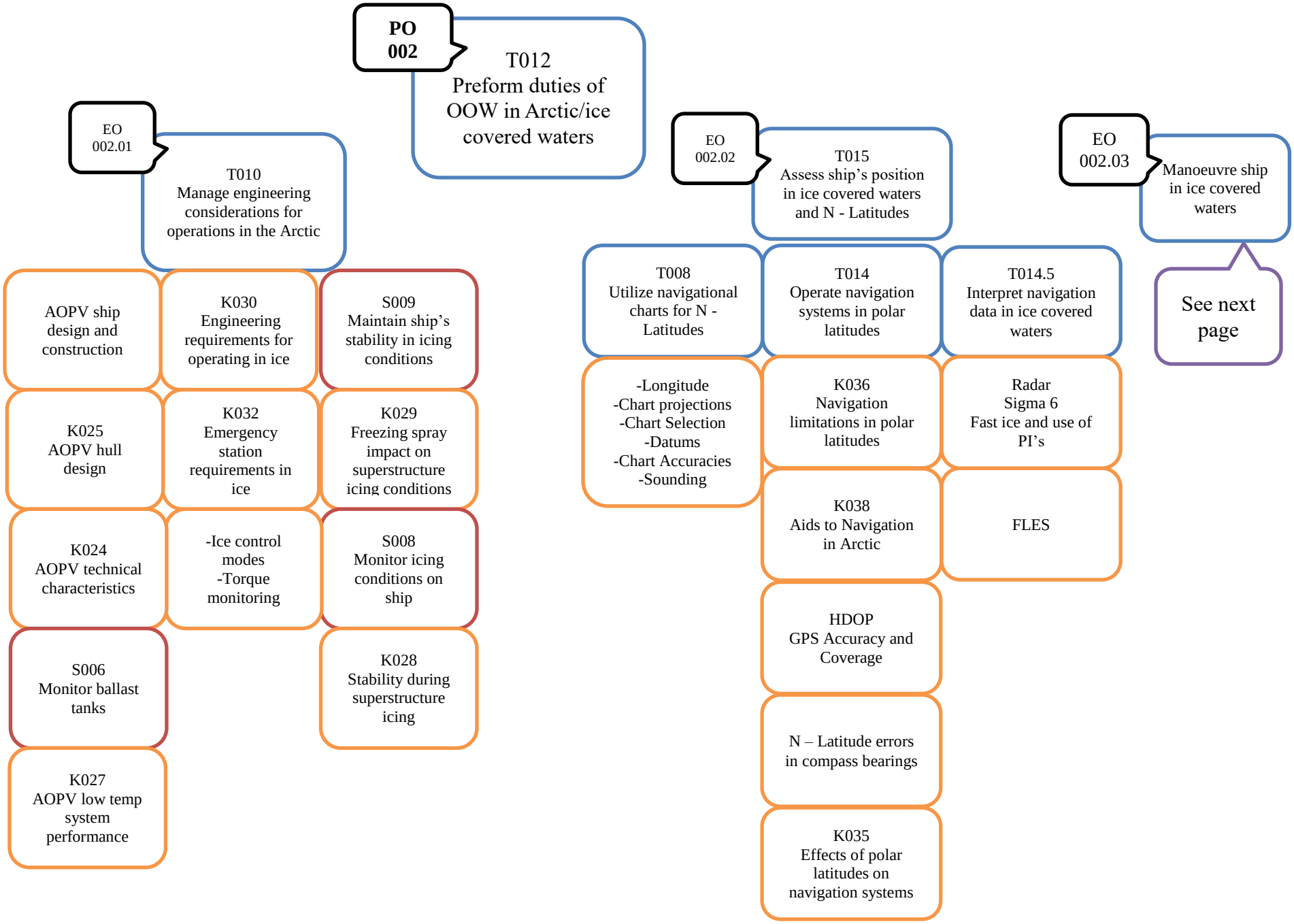
Legend:	
Task:	
Skill:	
Knowledge:	
PO / EO:	



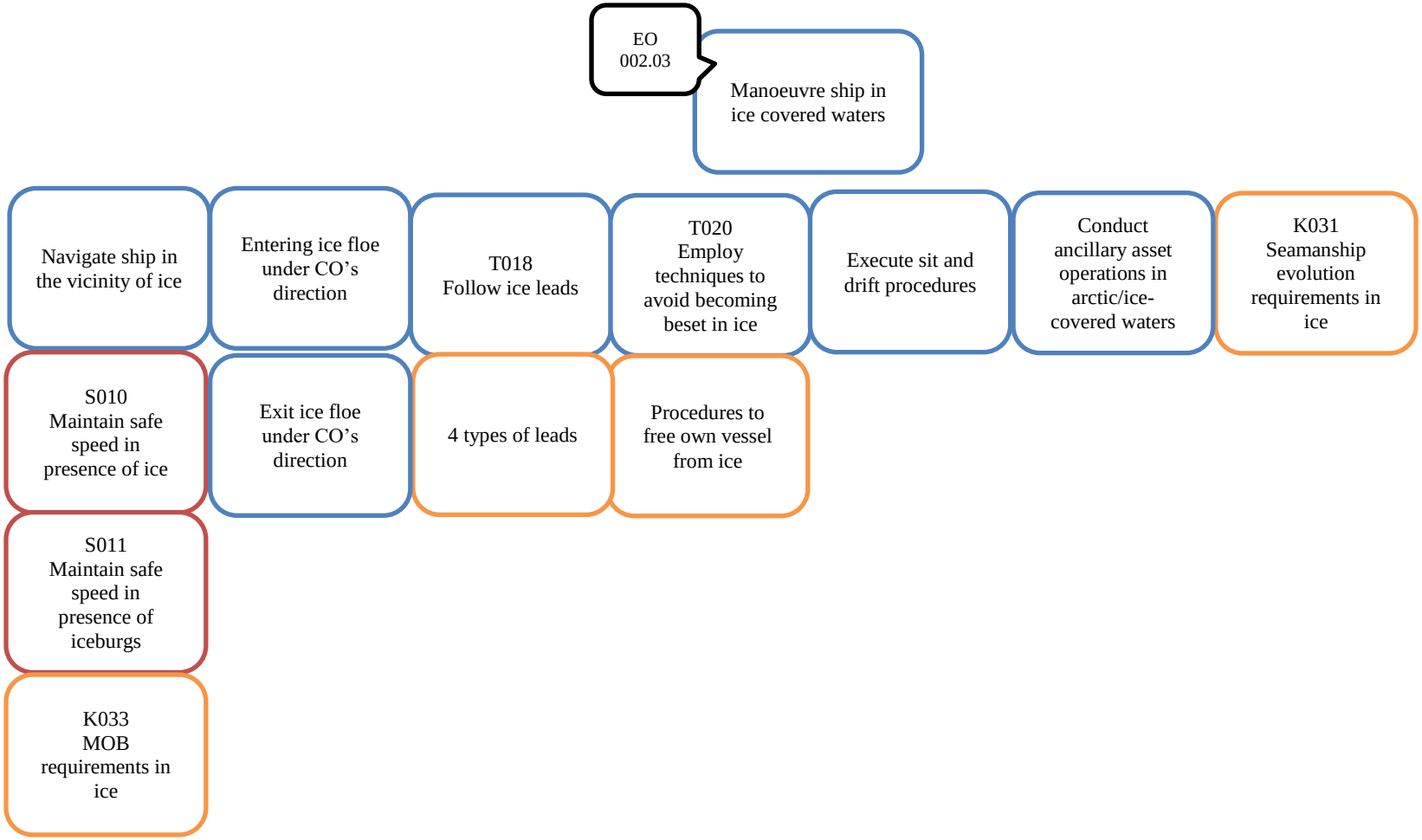
MARPAC RDIMS #1087689
Chap 1 approved DD MM YYYY
Chap 2, 3 approved DD MM YYYY



MARPAC RDIMS #1087689
Chap 1 approved DD MM YYYY
Chap 2, 3 approved DD MM YYYY



MARPAC RDIMS #1087689
Chap 1 approved DD MM YYYY
Chap 2, 3 approved DD MM YYYY


EO
002.03
Manoeuvre ship in ice covered waters
Navigate ship in the vicinity of ice
Entering ice floe under CO's direction
T018
Follow ice leads
T020
Employ techniques to avoid becoming beset in ice
Execute sit and drift procedures
Conduct ancillary asset operations in arctic/ice-covered waters
K031
Seamanship evolution requirements in ice
S010
Maintain safe speed in presence of ice
Exit ice floe under CO's direction
4 types of leads
Procedures to free own vessel from ice
S011
Maintain safe speed in presence of iceburgs
K033
MOB requirements in ice

ANNEX E - *ON JOB PERFORMANCE RECORD (OJPR)*

There is no requirement for an OJPR for the Basic Arctic Navigation Course therefore this Annex has been left blank.

ANNEX F - *ACRONYMS AND ABBREVIATIONS*

Acronym or Abbreviation	Term or Definition
AAN	Advanced Arctic Navigation Course
AIRSS	Artic Ice Regime Shipping System
AOPV	Arctic and Offshore Patrol Vessel
ASPPR	Arctic Shipping Pollution Prevention Regulations
ATT	Actual Training Time
BAN	Basic Arctic Navigation Course
BAT	Base Administrative Time
CAC	Canadian Arctic Category
CCG	Canadian Coast Guard
CFITES	Canadian Forces Individual Training and Education System
CIS	Canadian Ice Services
CNPTG	Commander Naval Personnel and Training Group
CO	Commanding Officer
DLN	Defence Learning Network
DND	Department of National Defense
EC	Enabling Check
ECAREG	Eastern Canada Vessel Traffic Service System
ECCC	Environment and Climate Change Canada
ECPINS	Electronic Charting and Position Indicating System
EDI	Explain Demonstrate Imitate
ENC	Electronic nautical chart
EO	Enabling Objective
F2F	Face to face
FLES	Forward Looking Echo Sounder
FNO	Fleet Navigating Officer
GPS	Global Positioning System
HDW	Harry DeWolf
IACS	International Association of Classification Societies
IAW	In Accordance With
IMO	International Maritime Organization
IVO	In vicinity of
JTFN	Joint Task Force North
LMS	Learning Management System
LOSV	Light Over Snow Vehicle
LRH	Leadership, Respect, and Honour

MANICE	Manual of Ice Navigation
MARPOL	The International Convention for the Prevention of Pollution from Ships
MCTS	Maritime Communications and Traffic Services
METOC	Meteorological and Oceanographic Centre
MilCOTS	Military Commercial Off the Shelf
MITE	Military Individual Training and Education
NABS	Navigation and Bridge Simulator
NFP	Notice for Power
NFR	Naval Force Readiness
NFS (A)	Naval Fleet School Atlantic
NFS (P)	Naval Fleet School Pacific
NLT	No Less than
NNF	Nanisivik Naval Facility
NORDREG	Northern Canada Vessel Traffic Service System
NPTG	Naval Personnel and Training Group
NQual	National Qualification
NTDC(P)	Naval Training Development Centre Pacific
NWO	Naval Warfare Officer
OGD	Other Government Department
OOW	Officer of the Watch
OTH	Over the Horizon
PC	Performance Check
PLAR	Prior Learning Assessment and Recognition
PO	Performance Objective
PPE	Personal Protective Equipment
PT	Physical Training
QSP	Qualification Standard and Plan
RACON	Radar Beacon
RCN	Royal Canadian Navy
RFI	Request for Information
SCTW	The International Convention on Standards of Training, Certification and Watchkeeping for Seafarers
SOLAS	International Convention for the Safety of Life at Sea
TAT	Training Administrative Time
TE	Training Establishment
TEL	Technology Enabled Learning
TRB	Training Review Board
TTT	Travel training Time
UAS	Unmanned Aircraft System
VTC	Video Tele-Conference

ANNEX G - *SUGGESTED SEQUENCING AND TIMINGS*

1. This annex is to be used by Campus Support to properly schedule all required classrooms, trainers, ships, personnel, etc for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).
2. The table is completed in the preferred sequence for lessons and resources. The embedded Excel document can be used as a template for any required amendments.

Remarks:

1. Course should be scheduled in the spring whenever possible, prior to the Arctic's navigable season.
2. On the west coast, it would be beneficial to have BAN scheduled immediately after the FNO course to facilitate FNO graduates who are posted to arctic-bound ships to take the course.
3. The suggested course sequencing is based on the need to deliver the course in a F2F manner. If there is no need to deliver the course F2F, the students will complete the DLN course which will follow a similar structure.

Annex G_Sequencing and t

EO ###.##	EO Performance Statement	Personnel resources	Required Resources (for scheduling)	Trg Time per resource (mins)	EC Time (mins)	PC Time (mins)
PO 001: Assess Impact of Environmental Conditions in ice and at Northern Latitudes						
PO ATT:	**910**	NOTES: Panel SMEs can be scheduled concurrently or separately depending on their availability				
EO 001.01	Apply Acts and Regulations that Govern Arctic Operations	1 Instructor / 6 trainees	Classroom w/ DWAN	90		

		Panel/ SME MCTS	Room with VTC capable of connecting with OGDs or, if SMEs available in person, classroom	30		
		Panel/ SME CCG		50		
		Panel/ SME ECCC		30		
		Panel/ SME JTFN Ops		15		
		1 Instructor / 6 trainees	Classroom w/ DWAN		15	
EO 001.02	Assess Ice Conditions	1 Instructor / 6 trainees	Classroom w/ DWAN	260		
			NPTT	70		
			Classroom w/ DWAN	140		
EO 001.03	Interpret Meteorological Data for the Canadian Arctic	SME	Classroom w/ DWAN	90		120
PO 002: Perform Duties of OOW in Arctic / Ice-Covered Waters						
PO ATT:	**1405**	NOTES:				
EO 002.01	Manage Engineering Considerations for Operations in the Arctic	AOPV Engineering Officer	AOPV ship tour	120		
			Transport	30		
		1 Instructor / 6 trainees	Classroom w/ DWAN		120	

EO 002.02	Assess Ship's Position in Ice-Covered Waters and Northern Latitudes	1 Instructor / 6 trainees	Classroom w/ DWAN	120		
					30	
EO 002.03	Manoeuvre the Ship in Ice-Covered Waters	1 Instructor / 6 trainees	Classroom w/ DWAN and video projecter	135		
			NABS	150		
			Classroom w/ DWAN and video projecter	100		
			Classroom w/ DWAN		210	
			NABS			390
		Actual trg time:	**2315**	**Total mins:**		**2435**
PT:	**0**	**Course admin time:**	**120**	**Total days:**		**6.1**

ANNEX H - *ADDITIONAL TRAINING RESOURCES*

All resources currently listed in paras 3-4 of Ch. 1.

(**Annex H not to be deleted**)

ANNEX I - *COSTING INFORMATION*

Include information about cost in whichever format is applicable (**Not to be deleted**).

Template Annex I Costing.xls

ANNEX J - *COURSE HISTORY*

This annex shall NOT be deleted. It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Reason for amendment (with date)	Details on course history
QSPWB 23 Apr – 2 May 18	QSPWB was held to support the Speciality qualification (Basic Arctic Navigation ALIB) for NWOs. The tasks, knowledges, and skills were taken from the specialty specification. They were compared to elements of the existing Arctic Operations Course MITE 119054) which was embedded into the Fleet Navigating Officer course; ALIB replaces that qualification. The specialty qualification was created based on new RCN capabilities with the introduction of the Arctic and Offshore Patrol Vessels, however this training is geared toward NWOs posted to any platform of ship that is venturing to the Arctic or ice covered waters.
QSP Review for DL	-Written directive embedded -Explain incorporation of DL, reasons, impact, actions, etc.

ANNEX K - *ASSESSMENT PLAN*

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
EC 001.01	EO 001.01	Apply acts and regulations that govern arctic operations	Using DLN, trainees will complete, on their own time, a 15-20 question test pertaining to the EO material.	70%	Remedial Training / Re-test
PC 001	PO 001	Assess the impact of environmental conditions at northern latitudes	Given a unique scenario, trainees will deliver a decision brief explaining their recommended COA based on their assessment of environmental conditions.	70%	Remedial Training / Re-test with new scenario
EC 002.01	EO 002.01	Manage engineering considerations for operations in the arctic	Trainees will deliver a brief comparing the engineering considerations of the HDW class to other ice capable ships.	SAT on all checklist items	Remedial Training / Re-test
EC 002.02	EO 002.02	Assess ship's position in ice-covered waters and northern latitudes	Trainees will complete, on their own time, a 10-15 question written test pertaining to the EO material.	Pass/Fail	Remedial Training / Re-test
EC 002.03	EO 002.03	Manoeuvre the ship in ice-covered waters	Given scenarios, trainees will act as OOW and navigate ice-covered waters in NABS	SAT on all Critical; 70% SATs overall	Remedial Training / Re-test
PC 002	PO 002	Perform the duties of OOW in arctic or ice-covered waters	Using the same scenarios given for PC 001, trainees will act as an OOW and navigate ice-covered arctic waters in NABS.	SAT on all Critical; 70% SATs overall	Failure to achieve ALIB / MITE

ANNEX L - *TEL PLAN*

EO	Recommended TEL Delivery	Interim Delivery	**Impact of TEL** - Realized benefits
001.01	DLN LMS EC with automatic correction/feedback online	Paper EC	**Effectiveness**: DLN LMS written exams are an equally effective means of disseminating an exam. Automatic feedback improves retention as trainees have more time to review incorrect answers without peer/instructor input. **Efficiency**: Paper ECs unnecessarily use extra resources (ink, paper, etc). The LMS will automatically correct the ECs leaving additional time for the instructors.
001.02	DLN LMS Homework assignment	Paper worksheet	**Effectiveness**: Automatic feedback improves retention as trainees have more time to review incorrect answers without peer/instructor input. Quality of pictures is improved online (limits reproduction inconsistencies) making the match assignment more realistic. **Efficiency**: Paper worksheets unnecessarily use extra resources (ink, paper, etc). The LMS will automatically correct the ECs leaving additional time for the instructors.
001.02	DLN LMS Module	F2F instruction	**Effectiveness**: DLN LMS interactions would be as effective as F2F instruction. **Efficiency**: Training can be used for BAN, AAN, and Met Tech self-study module. DLN LMS training can be used as quick review on the job.
002.03	Short animated videos: Berthing procedures in ice covered waters Slipping procedures in ice covered waters	Still pictures	**Effectiveness**: Still pictures are not as realistic; short videos will add realism to the material. Different angles which are hard to reproduce by pictures can be shown, including events happening below the water/ice surface. Standardized procedures can be demonstrated ensuring compliance with SSOs and negating instructor subjectivity.

	Conducting ancillary asset operations in arctic/ice covered waters		**Efficiency**: Videos can be watched as often as desired on the trainees own time to better understand the material without taking additional class / instructor time. Videos can be used as quick review onboard ship prior to berthing/slipping.

ANNEX M - *TARGET POPULATION AND TRAINING ESTABLISHMENT ANALYSIS*

The following information was compiled during the QSPWB (Apr 2018) to give context for developers. It can be removed from the QSP if desired after initial course development.

Key Target Population Details – BAN candidates (Post-NWO Phase IV)

"The learners' level of skill and knowledge with respect to the performance objectives (POs) is a factor which depends mainly on previous related training or education, and experience. It may also be affected by strong opinions or attitudes about the subject matter. Information should be obtained on the learners" -CFITES Vol 4

Characteristic	Target population details
General data	
• Class size • Expected annual throughput; • Location(s) of the intended learners; • Access to computers. • Gender • Age • Trades	• 6-12 per course; • 24 NWO total: 12 per coast • East and West coast (fleet) • Yes • 20% female • 25-35 • NWO
Learning skills and preferences	
Competencies	Situational awareness; perceptual processing speed; spatial orientation; decision-making; cognitive ability; working memory;
Reading and writing level	University-level
Preferences, attitudes towards specific type of learning material and equipment	Preference for TEL; doing versus sitting in class.
Previous military training experience	NWO phase training
Current subject-matter competence	
Experience with target performance	NWO: none

Characteristic	Target population details
Ability to perform any part of the POs	Core skills already; arctic and ice requirements are new
Perception of the impact on self, work or career	Different enviro; confidence builder; Posting-related; highly motivated to succeed
Job/career stage	Early for NWO
Social Characteristics	
Relationship to peers	Familiar with peers (likely shipmates)
Tendencies toward cooperation or competition	Cooperation

<u>Describing the learning environment</u> – LMS

The "learning system" is composed of all factors that affect and are affected by the learning that takes place: learners (Target Population), instructional materials, the instructors, instructional equipment (such as computers, projectors, LMS, etc), the instructional facilities, and the community or organization.

1. What are the characteristics of the teachers/trainers who will be using these materials?
What are the interests and preferences of the instructors? FNI (bare minimum Navigation background) *How do instructors feel about having instruction delivered via media or other nontraditional methods?* Good with TEL *What experience have instructors had with technology-based instruction?* DLI
2. Are there existing courses/programs into which this piece of instruction must fit?
If so, what is the strategy used in these materials?
3. What hardware is commonly available in the potential learning environments?
Are computer workstations available? How many? DWAN, standalone? DLN LMS *What software or other materials are available?*

4. What are the characteristics of the classes and facilities that will use the new instruction?